

Numerical and experimental studies on scattered mechanical properties for 3D needled C/C-SiC composites

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ABSTRACT

3D needled C/C-SiC composites show scattered mechanical properties in the experimental tests due to the random needling region distributions in the composites and the variability of microstructures at the needling regions. A specified methodology is developed to predict the scattered mechanical properties of the composites. Meso-scale models of the test specimens are established, in which the random distribution of needling regions and the uncertain properties of needling regions are considered. The predicted coefficients of variation for initial modulus, strength and failure strains of the composite agree well with the experimental results. Size effect of the test specimens on the experimental results is well analyzed by the proposed method. The influence of needling density, depth and distribution on the uncertainty of composite properties are also obtained. The methodology introduced by this work would guide the design of the proper specimen size and manufacturing needled materials with lower scattered mechanical properties.

1. Introduction

Three dimensional needle-punching (or 3D needling) technology [1–6] has been used to produce 3D fiber reinforcements, in which the in-plane fibers of the fabric plies can be transferred to the thickness direction by needling process to enhance the delamination resistance. 3D needling technology is attracting growing interest due to its simple process and low cost. Compared with the traditional 3D preform forming technologies, such as stitching, weaving, braiding and knitting, 3D needling technology shows higher cost-effectiveness [7]. At present, 3D needled composites (including C/C and C/C-SiC) have been widely used in aeronautics and space industries due to their superior mechanical and thermal properties.

The microstructure and mechanical properties of 3D needled composites have been widely studied. Failure modes and strength of the composites under different loading conditions have been investigated [8–13]. The experimental results indicated that 3D needled composites exhibited significant nonlinear deformations before collapse. What's more, the strength, elastic modulus and failure strain of the composites all showed some scattered. Tensile strength of the composite can have a coefficient of variation (CoV) of up to 13.22%. The scattered mechanical properties would reduce the safety factor of the composite

structures and limit the application of this family of materials seriously. Uncertainties are generally included in the mechanical properties for fibrous composites [14,15]. Weibull distribution functions [16,17] are usually employed to describe the statistic distributions of fibers, fiber bundles and fibrous composites. Calard et al. [18,19] studied the statistical distribution of the ceramic-matrix composites and proposed a macroscopic probabilistic model to predict the ultimate failures.

Meso-scale approaches [20,21] were used to research the deformation, stress distribution of 3D composites. The interface debonding behavior as well as the damage of yarns and matrix can be captured by finite element method (FEM). The stiffness and strength properties of the composites are derived from the calculated stress–strain curves under typical loading cases. Micro- and meso-scale approaches [22–25] combined with the finite element method (FEM) were also performed to analyze the scattered mechanical properties of the composites. In these models, the variabilities occurring at different scales were considered, such as the uncertain strength of fibers, variation of yarn path and cross-sections. 3D inclusions and voids in the material can also be modeled within the framework of FEM [26,27].

For 3D needled composites, the equivalent properties cannot be easily obtained by the unit cells, because the fiber architectures of the composites are difficult to be constructed explicitly in the meso-scale.

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Up to present, some useful attempts [28–31] have been made and several meso-scale models were established based on the fiber architectures of the needled composites. However, these models failed to analyze the uncertainty of the composite properties thoroughly. The CoVs of mechanical properties for needled composites can only be obtained by experiments at present, which was costly and time-consuming.

In this paper, the scattered mechanical properties of 3D needled C/C-SiC composites are studied. A novel meso-scale FE model is proposed to predict the stiffness and strength properties of the composites. The different needling region distributions of the specimens are explicitly constructed in the FE models. The uncertainty of the needling region properties are particularly considered by defining mechanical properties of each needling region with random functions. CoVs of tensile and shear properties of the composites can be numerically determined. Fairly good agreement is achieved between the CoVs of the predicted and experimental results. Furthermore, the influence of specimen size and needling parameters on the statistical characteristics of the composite properties are investigated.

2. Experimental investigation

2.1. Materials

The 3D needled C/C-SiC composites were prepared using the needled carbon fiber preforms by the CVI (chemical vapor infiltration), PIP (polymer impregnation-pyrolysis) and LSI (liquid silicon infiltration) process [32,33]. The 3D needled preform was prepared by the non-woven cloth and short-cut fiber net plies as shown in Fig. 1. Firstly, 0°, 90° non-woven cloth and the fiber net plies were stacked together alternatively. Secondly, the fibrous plies were needled on the surface by the needles. By repeating the two procedures, the needled preforms with a certain thickness and needling density can be produced. In this paper, the 0° and 90° fiber directions are referred to as x and y directions. The mechanical properties along x and y directions were supposed to be identical due to the symmetry of the fiber structures. During the needling process, some in-plane fibers could be cut off and some would be transferred to the thickness direction (z direction). A needling hole would be formed after the needle pull out from the fibrous plies. As the preform was needled repeatedly, the distributions of the needling holes are dense and disordered. Some regions could be needled for more than once. The voids can be observed in the needling regions of the

composite due to the uncomplete densification of the needling holes. The needling regions are supposed to be the defects of the material and cracks would always initiate from these regions during the tests. The fiber architectures and void contents for different needling regions are of great variability. As a result, the mechanical properties of the needling regions would exhibit uncertain.

2.2. Scattered mechanical properties

Two groups of tensile specimens and one group of shear specimens of 3D needled C/C-SiC composite (with the needling density 24.96 punch/cm² and needling depth 5 mm) were prepared and tested. The schematics of the specimens are shown in Fig. 2. The tensile and shear strains were measured by the strain gages mounted at the center position of the specimens. Both the tensile and shear tests were conducted on an Instron-1186 test machine with a crosshead loading speed of 50 MPa/min.

The experimental results of the tensile and shear specimens are provided in Fig. 3. The typical stress-strain curves obtained from tension and shear tests exhibit significant nonlinear characteristics before collapse. The ultimate failure stresses and strains show some scattered. The average values (AVGs) of the experimental data are summarized in Table 1. The coefficient of variation (CoV, written as Eq. (1)) for the experimental results are also provided to evaluate the scatter of the composite properties

$$\text{CoV} = S/\bar{x} \times 100\% \quad (1)$$

where S and $\bar{x}$ are standard deviation and AVG of the experimental results.

The scatter of the composite mechanical properties are related to the heterogeneity and complexity of the needled fiber reinforcements. The stress state and strength for the local regions of the composite would be of great variability in the tests. In this work, it is supposed that the scattered mechanical properties of 3D needled C/C-SiC composites are mainly induced by two reasons: (a) The random distribution and different volume contents of needling regions in the test specimens; (b) The uncertain properties of the needling regions caused by the variability of microstructures. The influence of other micro-scale defects (such as micro-cracks and voids in the matrix of un-needling regions) on the scatter of composite properties are neglected. It was supposed that these micro-scale defects distribute uniformly in the un-needling regions.

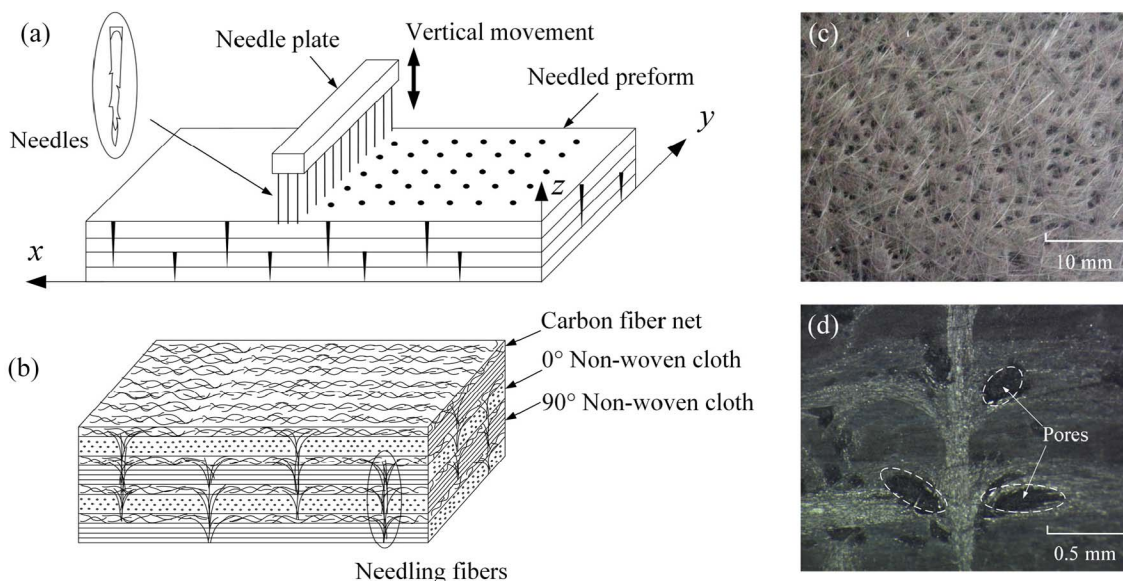


Fig. 1. (a) Schematic diagram of the needling process, (b) needling holes on the preform, (c) microstructure of the needled preform, (d) needling regions in the composite.

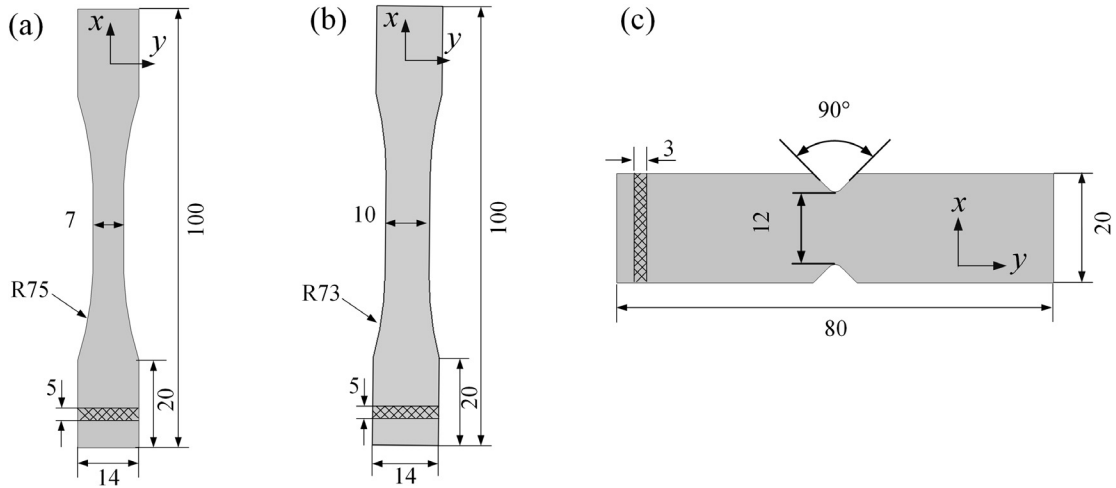


Fig. 2. Test specimens of 3D needled C/C-SiC composite. (a) 7 mm tensile specimen, (b) 10 mm tensile specimen and (c) shear specimen.

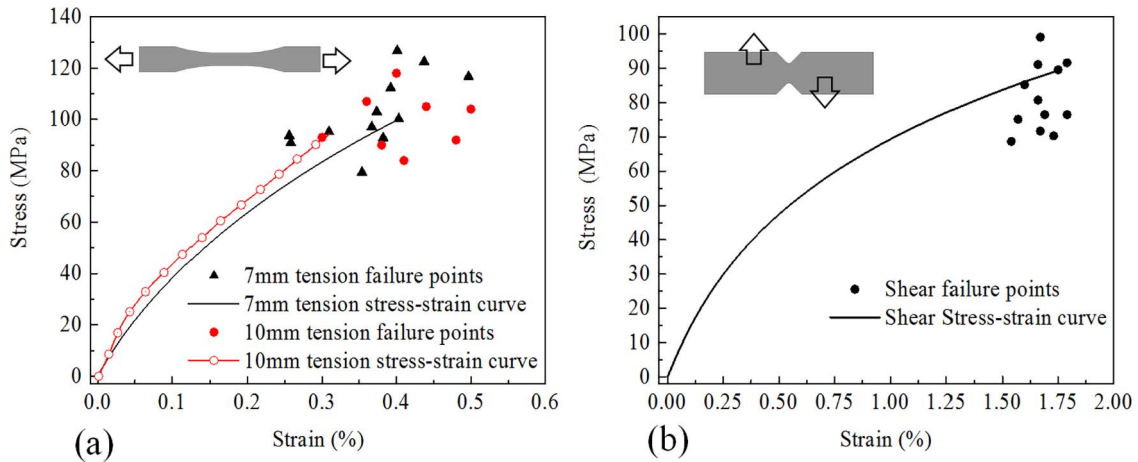


Fig. 3. The scattered experimental results of 3D needled C/C-SiC composites. (a) tensile tests, (b) shear tests.

Table 1

The experimental and FEM calculated properties of the composite.

		Modulus /GPa		Strength /MPa		Failure strain /%	
		AVG	CoV	AVG	CoV	AVG	CoV
7 mm	Tested	50.17	12.30%	102.63	13.22%	0.37	18.27%
	FEM	51.79	11.80%	104.22	12.74%	0.41	19.02%
10 mm	Tested	51.00	10.30%	99.13	10.56%	0.39	14.85%
	FEM	51.32	9.56%	103.91	11.18%	0.40	14.01%
Shear	Tested	12.99	9.65%	81.19	11.41%	1.68	4.81%
	FEM	13.47	8.73%	82.29	12.85%	1.66	5.08%

3. Numerical modeling for macroscopic properties

3.1. Finite element models of specimens

In this paper, the unit cell of 3D needled C/C-SiC composite (with needling density 24.96 punch/cm² and needling depth 5 mm) is established using the FE modeling method of Ref. [28]. The meso-scale models of the specimens can be constructed by selecting a part of the unit cell according to the specimen sizes, as shown in Fig. 4. During the mechanical tests, the strains were measured within the gauge length of the specimens, and the ultimate failure mostly occurred in these areas. Therefore, the meso-scale models only contain the center parts of the specimens to achieve a high computational efficiency. The *o*-xyz coordinate system was built in the unit cell model. Thus, the location of

tension and shear specimen models can be described by the coordinates (x_T, y_T, z_T) and (x_S, y_S, z_S), respectively. The distributions and volume contents of needling regions for the specimen models would be different if the specimen is selected from the different regions of the unit cell. The location coordinates of tension and shear specimen models are defined as Eqs. (2) and (3)

$$\begin{cases} x_T = i \frac{U_x - T_x}{m_x} & i = 0, 1, \dots, m_x \\ y_T = j \frac{U_y - T_y}{m_y} & j = 0, 1, \dots, m_y \\ z_T = k \frac{U_z - T_z}{m_z} & k = 0, 1, \dots, m_z \end{cases} \quad (2)$$

$$\begin{cases} x_S = i \frac{U_x - S_x}{n_x} & i = 0, 1, \dots, n_x \\ y_S = j \frac{U_y - S_y}{n_y} & j = 0, 1, \dots, n_y \\ z_S = k \frac{U_z - S_z}{n_z} & k = 0, 1, \dots, n_z \end{cases} \quad (3)$$

where the symbols T , S and U denote the dimensions of the unit cell, tension and shear specimen models respectively. In this paper, $U_x = U_y = 48$ mm, $U_z = 10$ mm; $T_x = 10$ mm, $T_y = 7$ mm (or 10 mm), $T_z = 5$ mm; and $S_x = 8$ mm, $S_y = 20$ mm, $S_z = 3$ mm. The parameters m and n are determined by Eqs. (4) and (5) to ensure each part of the unit cell can be selected ($\lceil \cdot \rceil$ is the round up symbol).

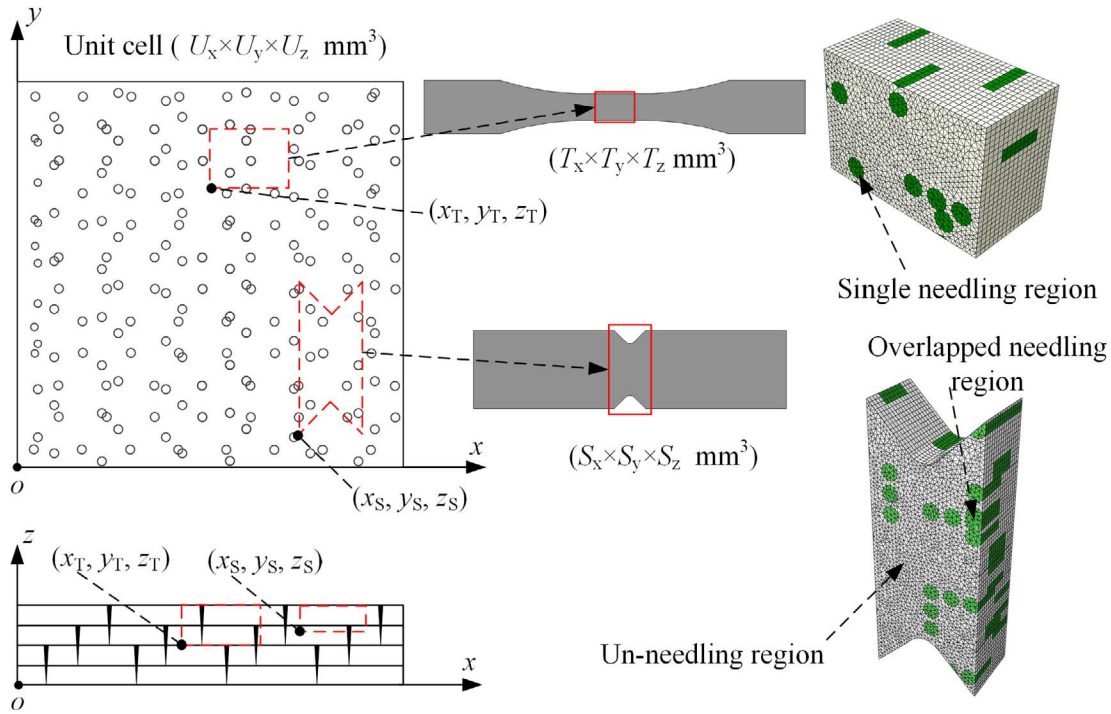


Fig. 4. Finite element models of the test specimens.

$$\begin{cases} m_x \geq [U_x/T_x] - 1 \\ m_y \geq [U_y/T_y] - 1 \\ m_z \geq [U_z/T_z] - 1 \end{cases} \quad (4)$$

$$\begin{cases} n_x \geq [U_x/S_x] - 1 \\ n_y \geq [U_y/S_y] - 1 \\ n_z \geq [U_z/S_z] - 1 \end{cases} \quad (5)$$

According to the dimensions of the test specimens, the parameters m and n are determined as $m_x = 4$, $m_y = 7$, $m_z = 2$, $n_x = 5$, $n_y = 4$ and $n_z = 3$. Finally, 120 FE models of 7 mm tension, 10 mm tension and shear specimens are selected from the unit cell respectively.

3.2. Material models of needling and un-needling regions

The specimen models are divided into the un-needling, single needling and overlapped needling regions. In-plane properties of the overlapped needling regions (being needled more than once) would be lower than that of the un-needling and single needling regions, as almost all the in-plane fibers of these regions were transferred to z direction. In this paper, the initial stiffness properties of un-needling and overlapped needling regions are defined by the parameters predicted by Ref. [28], as listed in Table 2. The stiffness parameters of single needling regions are considered to obey the uniform distribution, as expressed by Eq. (6)

$$\mathbf{S}_{ij}^S = f_{\text{uni}}(\mathbf{S}_{ij}^O, \mathbf{S}_{ij}^U) \quad (6)$$

where f_{uni} is the uniform distribution function; $\mathbf{S}_{ij}^U$, $\mathbf{S}_{ij}^O$ and $\mathbf{S}_{ij}^S$ are stiffness matrix of un-needling, overlapped needling and single needling

Table 2
The initial stiffness properties of un-needling and overlapped needling regions.

	E_x /GPa	E_y /GPa	G_{xy} /GPa	G_{xz} /GPa	ν_{xy}	ν_{xz}
Un-needling region	61.00	20.02	15.19	9.66	0.12	0.08
Overlapped needling region	15.65	91.99	6.28	10.21	0.25	0.23

regions.

Experimental investigations showed that, 3D needled C/C-SiC composites exhibited significant nonlinear deformations before collapse. The macroscopic nonlinear constitutive model of Ref. [10] was embedded into the FE models of the specimens to simulate the nonlinear stress-strain behavior of the composite. And the equivalent stress $\bar{\sigma}^*$, written as Eq. (7) [10], was used as the failure criterion of the specimen models

$$\bar{\sigma}^* = k^* \sqrt{a_1 \sigma_{11}^2 + a_2 \sigma_{22}^2 + \sigma_{12}^2} \quad (7)$$

where a_1 and a_2 are anisotropic parameters characterizing the anisotropic between normal and shear damage factors, k^* is the damage rate parameter. The failure index ϕ for un-needling, single needling and overlapped needling regions can be written as Eqs. (8)–(10) respectively.

$$\phi = \bar{\sigma}^*/\sigma_U = 1 \quad (8)$$

$$\phi = \bar{\sigma}^*/\sigma_S = 1 \quad (9)$$

$$\phi = \bar{\sigma}^*/\sigma_O = 1 \quad (10)$$

In which, σ_U , σ_S and σ_O represent the failure strength of the three regions mentioned above. σ_S is assumed to distribute uniformly between σ_U and σ_O

$$\sigma_S = f_{\text{uni}}(\sigma_O, \sigma_U) \quad (11)$$

Strength parameters σ_U and σ_O cannot be measured directly in the experimental tests. Nevertheless, the two parameters could be approximately estimated by the experiments. It was indicated by the tensile and shear tests that, the ultimate equivalent stress $\bar{\sigma}^*$ of the composite scattered from 60 to 85 MPa [10]. Therefore, the σ_U should be higher than 85 MPa. The σ_O could be much lower than 60 MPa, as most of the in-plane fibers of the overlapped needling regions were transferred to z direction. Four groups of parameters were adopted in the FEM calculations, as listed in Table 3. Suitable parameters can be determined by comparing the numerical and experimental results.

Table 3

Calculated results of 7 mm tensile specimen FE models with different values of the σ_0 and σ_U .

The values of σ_0 and σ_U	FEM calculated tensile strength	
	AVG/MPa	CoV/%
$\sigma_0 = 7.5$ MPa, $\sigma_U = 85$ MPa	100.12	8.84
$\sigma_0 = 5.0$ MPa, $\sigma_U = 85$ MPa	98.55	9.83
$\sigma_0 = 2.5$ MPa, $\sigma_U = 87.5$ MPa	104.22	12.74
$\sigma_0 = 1.0$ MPa, $\sigma_U = 90$ MPa	113.81	14.81

4. Results and discussions

4.1. Validation of predicted results

The initial stiffness properties of the specimens can be obtained by homogenization method. In the FEM calculations (conducted by Abaqus), periodic boundary conditions were applied to the FE models of the specimens. The FEM predicted elastic modulus of 7 mm tensile, 10 mm tensile and shear specimens are provided in Fig. 5(a), (b) and (c). The calculation results show some scattered and the scatter ranges are consistent with the experimental results. The AVGs and CoVs for the calculated modulus E_x and G_{xy} are summarized in Table 1. Fairly good agreement was achieved between the numerical and experimental results.

Uniaxial tension and in-plane shear loadings were applied to the tensile and shear specimen models. In the FE models, when the failure index ϕ exceeds 1.0, failure is assumed to occur and the elastic modulus E_x and G_{xy} would drop to zero. Four groups of parameters σ_0 and σ_U (listed in Table 3) were used to predict the strength of 7 mm tensile specimens. It was found that, if σ_0 and σ_U take the value of 2.5 and 87.5 MPa respectively, the numerical results could be very close to the experimental results. Hence, this group of parameters were adopted in the following calculations. The failure process of the specimen models are displayed in Fig. 6. During the loading process, failure first occurred at the needling regions, as the value of failure index ϕ for needling

regions are much higher than other regions in the model. The FEM predicted failure points and typical stress-strain curves of the specimen models are provided in Fig. 5(d), (e) and (f). It is indicated that the nonlinear deformation and ultimate failure of the composite can be reasonably described by the material model of this work. AVGs and CoVs of the strength and failure strains for the specimens are listed in Table 1. As we can see, the scattered composite properties are well predicted.

4.2. Influence of specimen size on the CoVs

The experimental results of fibrous composites are generally scattered. Experimental results of the specimens failed to provide the global composite properties but just reflect the local region properties of the composite, due to the limited specimen sizes. CoV of the experimental results can be reduced by conduct the tests with larger specimens. As shown in Table 1, the AVGs of the tensile properties obtained by 7 mm and 10 mm tensile specimens are consistent. However, CoVs of the mechanical properties obtained by 10 mm tensile specimens are lower than that obtained by 7 mm specimens.

FE models of the specimens with different sizes are established in this paper to analyze the influence of specimen size on the CoVs of composite properties, as displayed in Fig. 7. 120 FE models in each size were selected in the unit cell by Eqs. (2)–(5). The calculation results show that, CoVs of the mechanical properties would reduce dramatically with the increase of the FE model volume in a certain range. When the volume of tensile FE models increased from 350 mm³ to 1280 mm³, CoVs of tensile modulus (E_x), strength (σ_T) and failure strains (ϵ_T) dropped by 49.3%, 47.2% and 60.0%, respectively. Meanwhile, with the volume of shear FE models increased from 480 mm³ to 1920 mm³, CoVs of shear modulus (G_{xy}), strength (τ_S) and failure strains (γ_S) dropped by 42.6%, 61.9% and 54.7%, respectively.

The relationship between CoVs of mechanical properties and the volume of specimen FE models can be described by an exponential type fitting function (written as Eq. (12)), as seen in Fig. 8.

$$CoV = c_1 \exp(c_2/V) \quad (12)$$

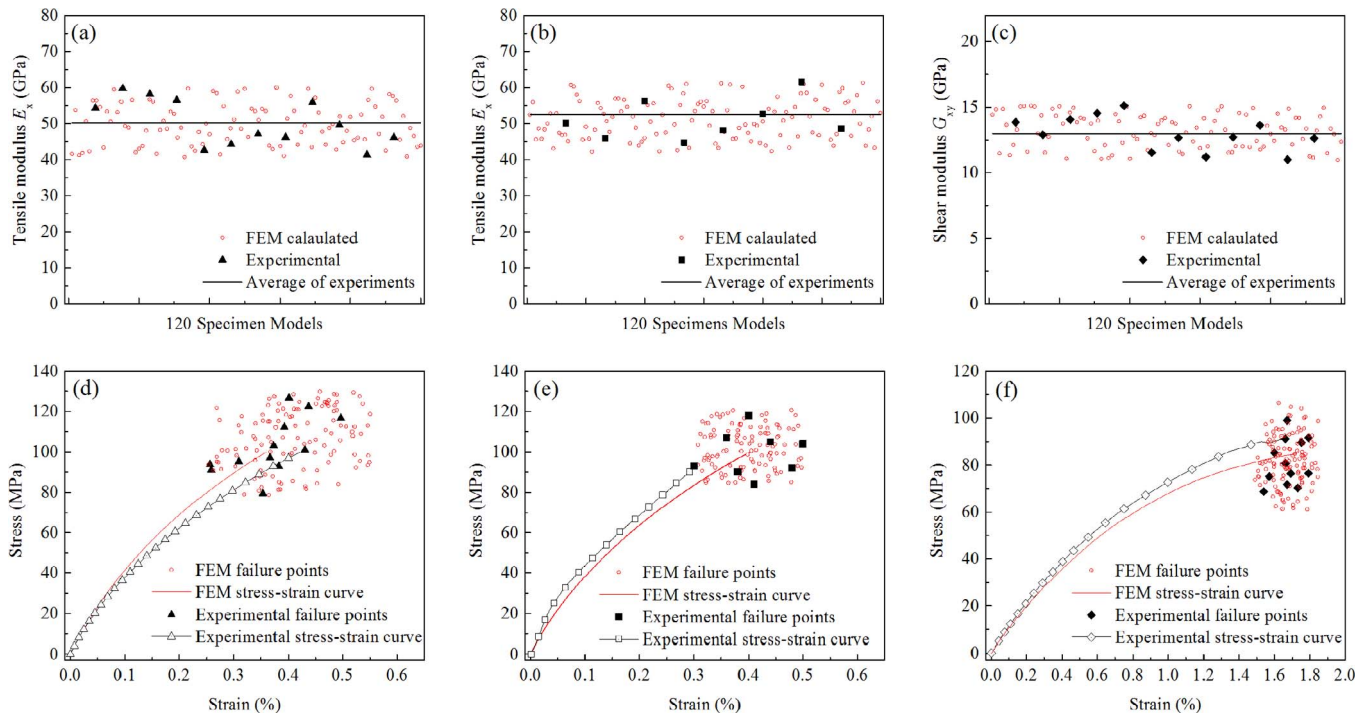


Fig. 5. Comparison between the FEM calculated and experimental results. (a), (b) and (c) elastic modulus of 7 mm tension, 10 mm tension and shear specimens; (d), (e) and (f) failure points of 7 mm tension, 10 mm tension and shear specimens.

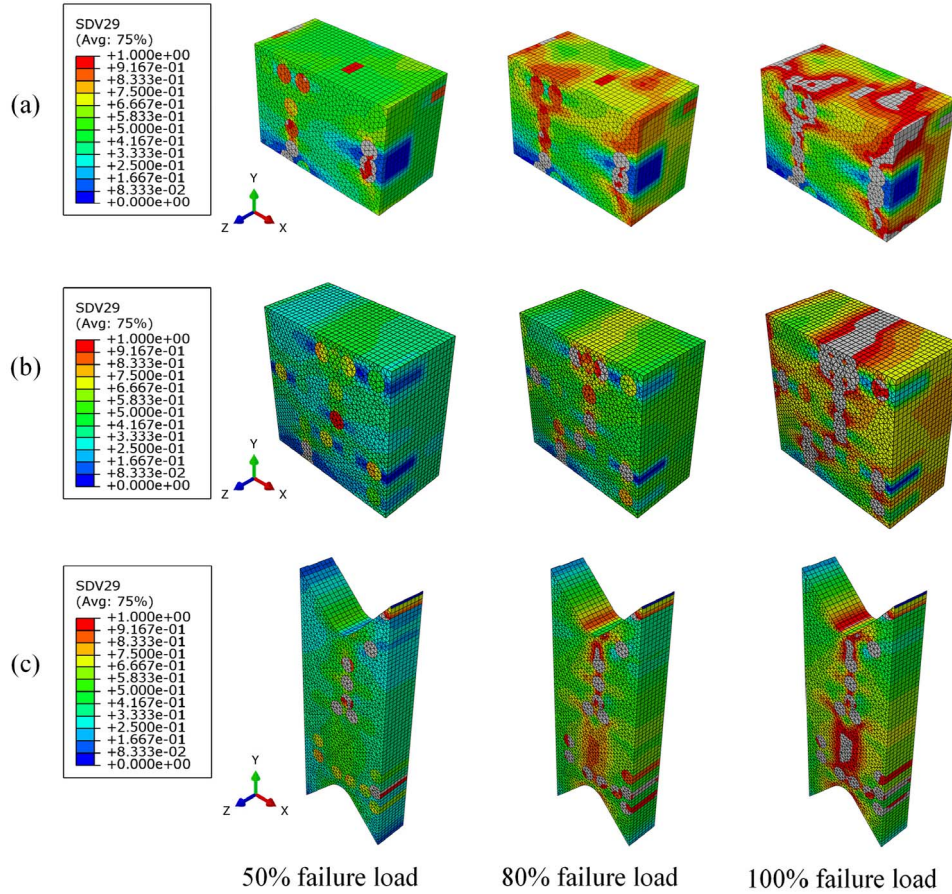


Fig. 6. Evolution of failure index ϕ for the specimen models. (a) 7 mm tensile specimen (b) 10 mm tensile specimen (c) shear specimen.

where V is the volume of the specimen FE models, c_1 and c_2 are fitting parameters. It should be noted that, after the volumes of tensile and shear FE models reached $14 \times 14 \times 5 \text{ mm}^3$ and $14 \times 35 \times 3 \text{ mm}^3$, the CoVs of tensile and shear properties tended to be stable and would not decrease significantly any longer. Thus, these two dimensions are

suggested to be adopted in future mechanical tests to achieve a balance between the reliability and economic efficiency of the experiments.

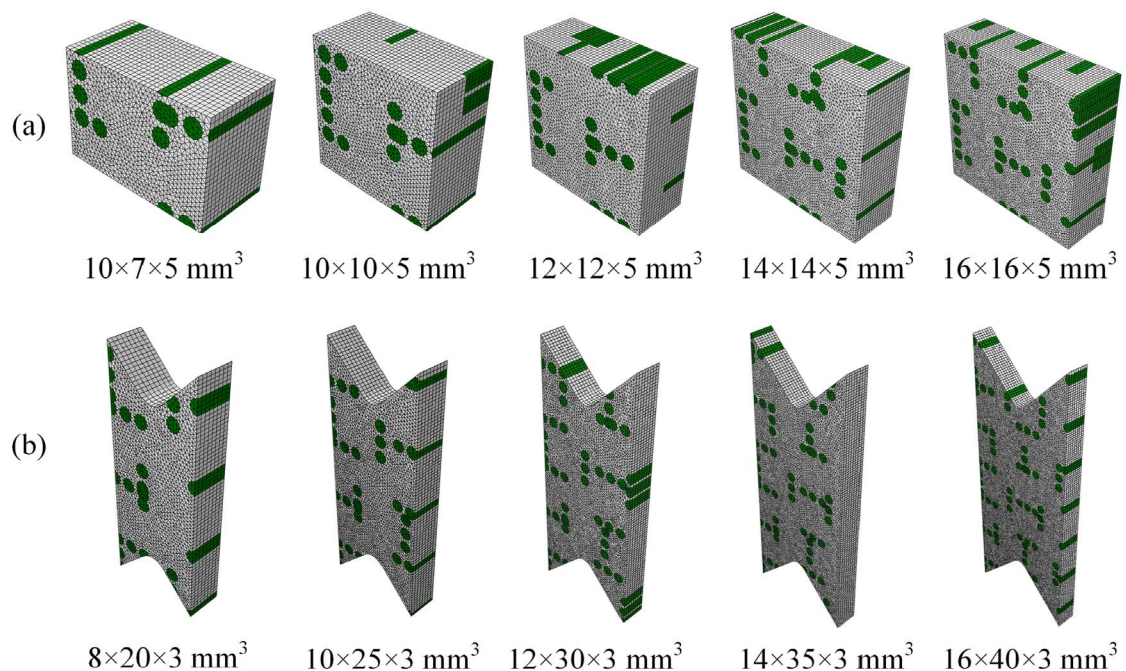


Fig. 7. The FE models of specimens with different sizes. (a) tensile specimen models (b) shear specimen models.

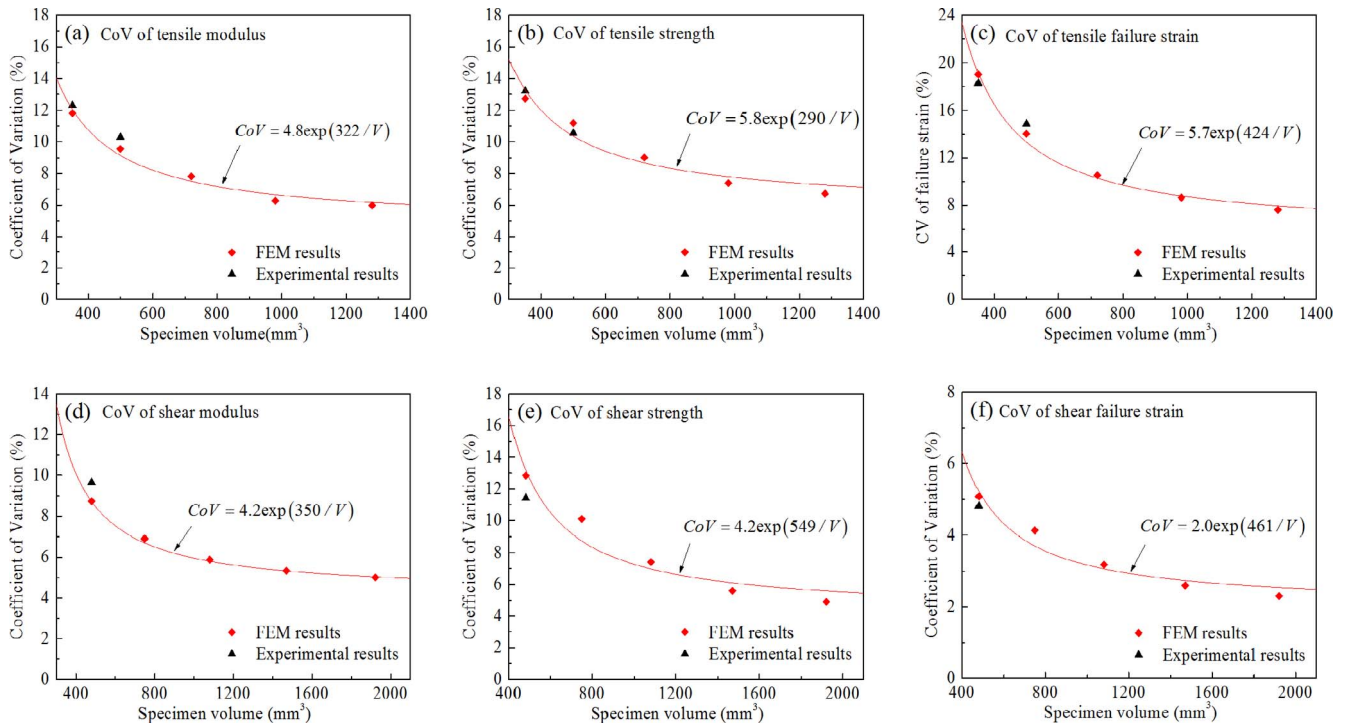


Fig. 8. The relationship between CoV of mechanical properties and the volume of specimen models. (a), (b), (c) CoVs of tensile modulus, strength and failure strains; (d), (e), (f) CoVs of shear modulus, strength and failure strains.

4.3. Influence of needling density on the CoVs

The mechanical properties of the needled composites are closely related to the needling parameters, such as needling density, depth and distribution [28,34–36]. In this paper, unit cells with five different needling densities (16.64, 20.8, 24.96, 29.12, and 33.28 punch/cm²) were established. The needling depth of the unit cells are the same (5mm), and the needling distributions tried to be uniform. From each unit cell, 120 tensile and shear FE models are selected by Eq. (2)–(5), respectively. The dimensions of tensile and shear FE models were set to be $10 \times 7 \times 5 \text{ mm}^3$ and $8 \times 20 \times 3 \text{ mm}^3$. Then the influence of needling density on the CoVs of the composite properties can be investigated by the five groups of FEM calculations.

Fig. 9 displays the relationship between the needling density and CoVs of mechanical properties for the composite. With the needling density increased from 16.64 to 32.28 puch/cm², CoVs of E_x and G_{xy} increased by 12.8% and 15.1%. The failure behavior of the composite is more sensitive to the needling density, because the variability of the strength is much more significant than that of initial modulus for the needling regions. The increasing ranges of CoVs for σ_T , ε_T , τ_S and γ_S are up to 88.3%, 53.9%, 61.4% and 64.1%, respectively, when needling density increased from 16.64 to 32.28 puch/cm². Interlaminar performance of the composites can be enhanced by increasing the needling density. However, the excessive needling density should be avoided in the realistic needling process. Because more defects and uncertain factors would be introduced into the composite if the needling density increases. That would result in the decrease of in-plane performance, and the CoVs of the in-plane properties could also be increased.

4.4. Influence of needling depth on the CoVs

The needling depth is another important parameter that influences the mechanical properties of the needled preforms and composites significantly [28,35–36]. Unit cells with the same needling density (24.96 puch/cm²), yet different needling depth (5, 7.5 and 10 mm) were established. From each unit cell, 120 tensile and shear FE models

(with the dimensions $10 \times 7 \times 5 \text{ mm}^3$ and $8 \times 20 \times 3 \text{ mm}^3$) were selected by Eq. (2)–(5), respectively.

In the needling process, more fibrous plies would be penetrated through with a larger needling depth. As a consequence, more needling regions were introduced to the unit cell models. The influence of needling depth on the CoVs of composite properties can be obtained by FEM calculations, as shown in Fig. 10. With the needling depth increased from 5 mm to 10 mm, CoVs of E_x , ε_T , G_{xy} and γ_S increased by 5.2%, 6.2%, 5.4% and 4.3%, respectively. The CoVs of the σ_T and τ_S are more sensitive to the increase of needling depth. When the needling depth increased from 5 mm to 10 mm, CoVs of σ_T and τ_S increased by 25.3% and 17.8%.

Therefore, the too large needling depth should not be adopted during the needling process in order to keep the CoVs of the in-plane properties for the needled composite at a relatively low level. It was also reported that, the interlaminar tensile and shear strength of the needled preforms will decrease if the needling depth is more than 12–14 mm [35]. Besides, the needles could be broken by the resistance of the fibrous plies if the needling depth is too larger.

4.5. Influence of needling distribution on the CoVs

The needling distributions in the preforms could be variability even with the same needling density and depth. Four different needling distributions with the same needling density (29.12 puch/cm²) were designed artificially, as displayed in Fig. 11 (referred to as distribution A, B, C and D). Then, four unit cells were established with the different needling distributions. The needling distribution A is relatively uniform, while the other three needling distributions are disordered. The volume contents of overlapped needling regions (V_0) in the unit cells corresponding to distribution A, B, C and D are 2.95%, 4.48%, 5.09% and 5.25%. 120 tensile and shear FE models (with the dimensions $10 \times 7 \times 5 \text{ mm}^3$ and $8 \times 20 \times 3 \text{ mm}^3$) were selected from each unit cell by Eq. (2)–(5), respectively.

The calculated results of the four groups of FE models are listed in Table 4. The mechanical properties of the composite are significantly

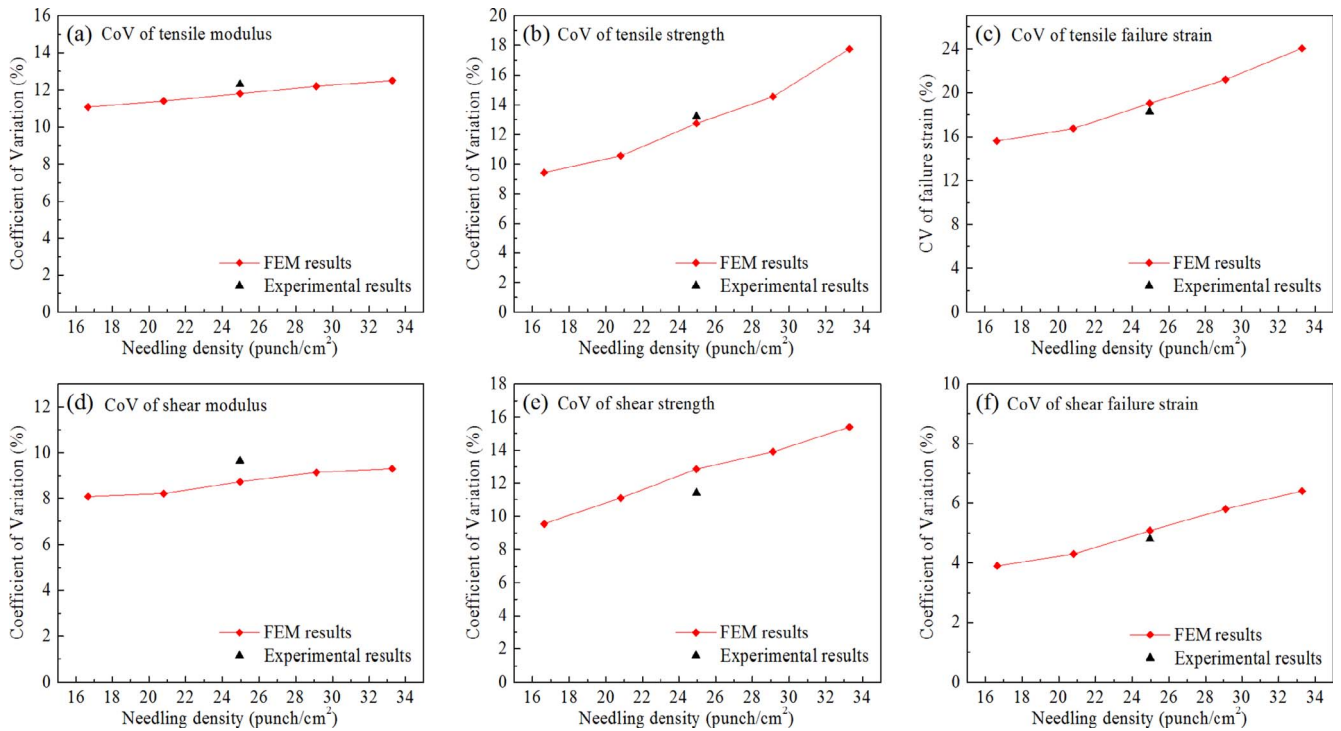


Fig. 9. The influence of needling density on the CoV of the composite properties. (a), (b), (c) CoVs of tensile modulus, strength and failure strains; (d), (e), (f) CoVs of shear modulus, strength and failure strains.

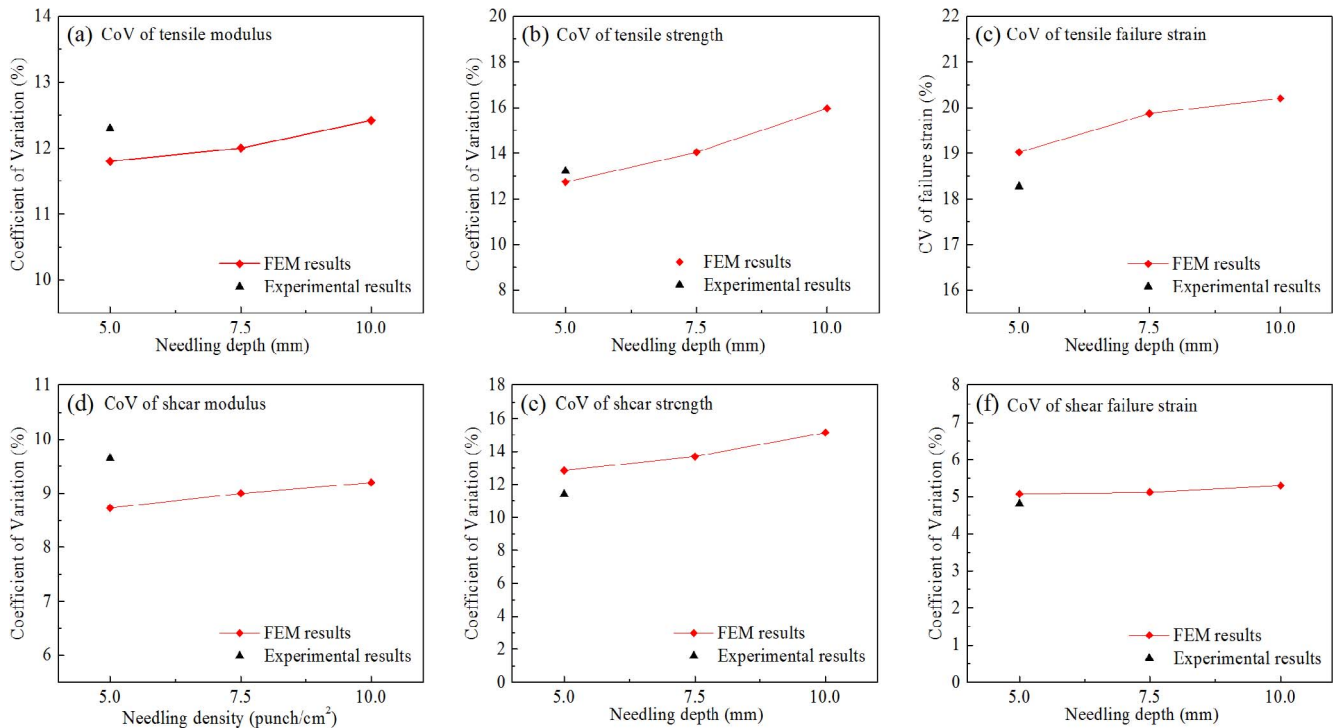


Fig. 10. The influence of needling depth on the CoV of the composite properties. (a), (b), (c) CoVs of tensile modulus, strength and failure strains; (d), (e), (f) CoVs of shear modulus, strength and failure strains.

influenced by the needling distributions. It is shown that, the average modulus (E_x , G_{xy}) and strength (σ_T , τ_S) decreased, while CoVs of these properties increased with the increase of V_0 . The average failure strains (ϵ_T , γ_S) for the four needling distributions have no obvious changes. However, CoVs for the failure strains slightly increased along with the increase of V_0 .

The content of overlapped needling regions V_0 could be increased, if

the needling distribution is more disordered. For overlapped needling regions, in-plane stiffness and strength properties are extremely low. Moreover, the overlapped needling regions distribute randomly in the composites. The in-plane performance of specimens with higher V_0 would surely be lower than the average level of the composite. As a result, the uniform needling distributions are requested in the needling process, to ensure the good in-plane performance and low variability of

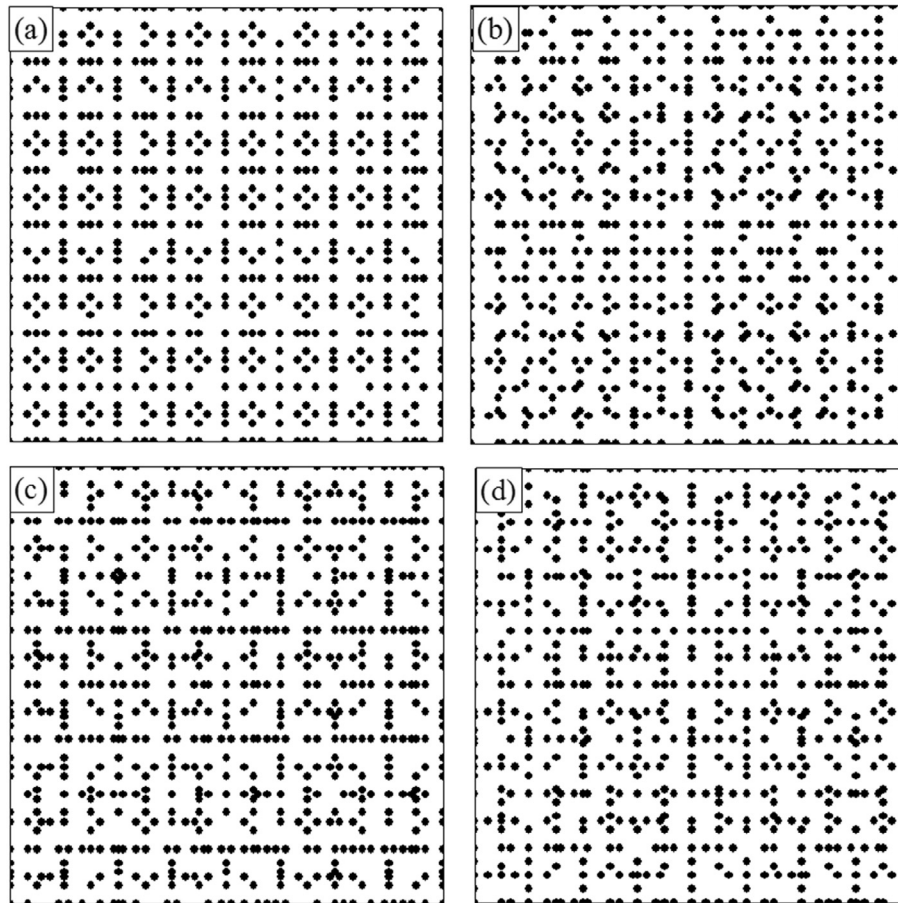


Fig. 11. Different needling distributions with the needling density of 29.12 punch/cm².

Table 4
FEM calculation results of the composites with different needling distributions.

	Distribution A		Distribution B		Distribution C		Distribution D	
	$V_0 = 2.95\%$ AVG	CoV /%	$V_0 = 4.48\%$ AVG	CoV /%	$V_0 = 5.09\%$ AVG	CoV /%	$V_0 = 5.25\%$ AVG	CoV /%
E_x /GPa	50.21	12.30	49.19	12.96	48.56	13.15	47.82	13.25
σ_T /MPa	95.67	14.54	92.42	15.9	91.62	16.01	89.12	16.51
ε_T /%	0.42	21.19	0.43	23.29	0.42	23.01	0.44	23.47
G_{xy} /GPa	13.30	9.94	13.01	11.96	13.15	12.18	12.87	12.25
τ_S /MPa	76.29	13.90	74.17	14.90	73.91	15.01	72.18	15.11
γ_S /%	1.69	6.20	1.71	7.46	1.70	7.50	1.72	7.60

the composite mechanical properties.

5. Conclusions

The mechanical properties of 3D needled C/C-SiC composites are investigated by tensile and shear tests. Meso-scale models of the specimens were established, both AVGs and CoVs of the mechanical properties for the composite were well predicted. The influence of specimen size and needling parameters on CoVs of the mechanical properties are quantitatively analyzed by the FEM calculations. With the increase of specimen volume, CoVs can be reduced effectively. More reasonable tensile and shear specimen sizes for further experiments are obtained. The excessive needling density and depth, as well as non-uniform needling distributions should be avoided in the needling process. The increase of volume contents for needling and overlapped needling regions would decrease the in-plane mechanical properties of the composite and make the macroscopic properties more scattered.

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